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# =====
# KNHANES Data Cleaning Script
# Depression Severity Modeling
# =====

library(tidyverse)
library(readr)

# -----
# 1. Load merged KNHANES data
# -----

DATA_PATH <- "KNHANES_even_years_2014_2024_merged.csv"

raw_df <- read_csv(DATA_PATH, show_col_types = FALSE)

cat("Original data shape:", dim(raw_df), "\n")

# -----
# 2. Define required variables
# -----

required_vars <- c(

  # ID / survey year
  "year",
  "ID",

  # Outcome: PHQ-9
  "mh_PHQ_S",

  # Demographics
  "age",
  "sex",

  # Socioeconomic covariates
  "ho_incm",
  "edu",
  "marri_1",
  "EC1_1",

  # Stress perception
  "BP1",

  # Sleep variables
  "BP8",
  "BP16_1",
  "BP16_2",
  "BP16_11",
  "BP16_12",
  "BP16_13",
  "BP16_14",
  "BP16_21",
  "BP16_22",
  "BP16_23",
  "BP16_24",

  # Physical activity variables
  "BE3_71", "BE3_72", "BE3_73", "BE3_74",
  "BE3_75", "BE3_76", "BE3_77", "BE3_78",
  "BE3_81", "BE3_82", "BE3_83", "BE3_84",
  "BE3_85", "BE3_86", "BE3_87", "BE3_88",
  "BE3_91", "BE3_92", "BE3_93", "BE3_94",

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# Resting pulse rate variables
"HE_PLS",
"HE_PLS_30",
"HE_mPLS",

# Body / behavior covariates
"HE_BMI",
"HE_obese",
"BS1_1",
"BS3_1",
"BD1",
"BD1_11",

# Disease history
"DI1_dg",    # 고혈압
"DI3_dg",    # 뇌졸중
"DI5_dg",    # 심근경색증
"DI6_dg"     # 협심증
)

existing_vars <- required_vars[required_vars %in% names(raw_df)]
missing_vars <- setdiff(required_vars, names(raw_df))

cat("\nNumber of required variables:", length(required_vars), "\n")
cat("Existing variables:", length(existing_vars), "\n")
cat("Missing variables:", length(missing_vars), "\n")

if (length(missing_vars) > 0) {
  cat("\nMissing variables:\n")
  print(missing_vars)
}

df_cohort <- raw_df %>%
  select(all_of(existing_vars))

cat("\ndf_cohort initial shape:", dim(df_cohort), "\n")

# -----
# 3. Helper functions
# -----

to_num <- function(x) {
  suppressWarnings(as.numeric(str_trim(as.character(x))))
}

valid_range <- function(x, lower, upper) {
  x <- to_num(x)
  if_else(x >= lower & x <= upper, x, NA_real_)
}

valid_values <- function(x, values) {
  x <- to_num(x)
  if_else(x %in% values, x, NA_real_)
}

remove_missing_codes <- function(x, missing_codes = c(88, 99)) {
  x <- to_num(x)
  if_else(x %in% missing_codes, NA_real_, x)
}

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# =====
# Exclusion criteria
# =====

# -----
# 4. PHQ-9 total score and 2-level severity group
# -----

df_cohort <- df_cohort %>%
  mutate(
    mh_PHQ_S = valid_range(mh_PHQ_S, 0, 27),

    PHQ_group = case_when(
      mh_PHQ_S >= 0 & mh_PHQ_S <= 4 ~ "Minimal",
      mh_PHQ_S >= 5 & mh_PHQ_S <= 27 ~ "Mild_or_greater",
      TRUE ~ NA_character_
    ),

    PHQ_group = factor(
      PHQ_group,
      levels = c("Minimal", "Mild_or_greater")
    )
  ) %>%
  drop_na(PHQ_group)

cat("\nPHQ-9 check after dropping missing:\n")
print(summary(df_cohort$mh_PHQ_S))
print(table(df_cohort$PHQ_group, useNA = "ifany"))
cat("df_cohort shape after PHQ drop:", dim(df_cohort), "\n")

# 44313 -> 32294
# n = 12019 excluded

# -----
# 5. Age
# -----

df_cohort <- df_cohort %>%
  mutate(
    age_numeric = valid_range(age, 0, 120)
  ) %>%
  filter(
    !is.na(age_numeric),
    age_numeric >= 19
  ) %>%
  select(-any_of("age"))

cat("\nAge check after adult restriction:\n")
print(summary(df_cohort$age_numeric))
cat("df_cohort shape after age cleaning:", dim(df_cohort), "\n")

# 32294 -> 32294
# n = 0 excluded

# -----
# 6. Resting pulse rate
# -----
# HE_mPLS = 60-second pulse
# HE_PLS_30 = 30-second pulse -> x2
# HE_PLS = 15-second pulse -> x4
#
# Priority:

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# 1. HE_mPLS
# 2. HE_PLS_30 * 2
# 3. HE_PLS * 4
#
# Exclusion:
# missing pulse
# pulse < 40 bpm
# pulse > 200 bpm

df_cohort <- df_cohort %>%
  mutate(

    HE_mPLS_numeric = if ("HE_mPLS" %in% names(.)) {
      to_num(HE_mPLS)
    } else {
      NA_real_
    },

    HE_PLS_30_numeric = if ("HE_PLS_30" %in% names(.)) {
      to_num(HE_PLS_30)
    } else {
      NA_real_
    },

    HE_PLS_numeric = if ("HE_PLS" %in% names(.)) {
      to_num(HE_PLS)
    } else {
      NA_real_
    },

    pulse_rate_60 = case_when(
      !is.na(HE_mPLS_numeric) ~ HE_mPLS_numeric,
      !is.na(HE_PLS_30_numeric) ~ HE_PLS_30_numeric * 2,
      !is.na(HE_PLS_numeric) ~ HE_PLS_numeric * 4,
      TRUE ~ NA_real_
    ),

    pulse_abnormal = !is.na(pulse_rate_60) &
      (pulse_rate_60 < 40 | pulse_rate_60 > 200)
  )

cat("\nPulse rate check:\n")
print(summary(df_cohort$pulse_rate_60))
print(table(df_cohort$pulse_abnormal, useNA = "ifany"))

cat("\nPulse rate missing/abnormal counts by year:\n")

print(
  df_cohort %>%
    group_by(year) %>%
    summarise(
      n = n(),
      pulse_missing = sum(is.na(pulse_rate_60)),
      pulse_abnormal_n = sum(pulse_abnormal, na.rm = TRUE),
      .groups = "drop"
    )
)

df_cohort <- df_cohort %>%
  select(
    -any_of(c(
      "HE_mPLS",
      "HE_PLS_30",
      "HE_PLS",
      "HE_mPLS_numeric",

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      "HE_PLS_30_numeric",
      "HE_PLS_numeric"
    ))
  )

n_before_pulse <- nrow(df_cohort)

df_cohort <- df_cohort %>%
  filter(
    !is.na(pulse_rate_60),
    pulse_abnormal == FALSE
  )

cat("\nResting pulse exclusion applied.\n")
cat("Before:", n_before_pulse, "\n")
cat("After:", nrow(df_cohort), "\n")
cat("Excluded:", n_before_pulse - nrow(df_cohort), "\n")

# 32294 -> 32093
# n = 201 excluded

# -----
# 7. Cardiovascular disease exclusion
# -----
# DI1_dg = hypertension
# DI3_dg = stroke
# DI5_dg = myocardial infarction
# DI6_dg = angina
#
# 0 = no
# 1 = yes
#
# Subjects with confirmed diagnosis are excluded.
# Unknown / nonresponse / unavailable-year values are retained.

df_cohort <- df_cohort %>%
  mutate(

    hypertension_dx = valid_values(DI1_dg, c(0, 1)),
    stroke_dx = valid_values(DI3_dg, c(0, 1)),
    mi_dx = valid_values(DI5_dg, c(0, 1)),
    angina_dx = valid_values(DI6_dg, c(0, 1)),

    cvd_exclusion = case_when(

      hypertension_dx == 1 |
        stroke_dx == 1 |
        mi_dx == 1 |
        angina_dx == 1 ~ 1,

      hypertension_dx == 0 &
        stroke_dx == 0 &
        mi_dx == 0 &
        angina_dx == 0 ~ 0,

      TRUE ~ NA_real_
    )
  ) %>%
  select(
    -any_of(c(
      "DI1_dg",
      "DI3_dg",
      "DI5_dg",
      "DI6_dg"
    ))
  )

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    ))
  )

cat("\nCVD variables check:\n")

print(table(df_cohort$hypertension_dx, useNA = "ifany"))
print(table(df_cohort$stroke_dx, useNA = "ifany"))
print(table(df_cohort$mi_dx, useNA = "ifany"))
print(table(df_cohort$angina_dx, useNA = "ifany"))
print(table(df_cohort$cvd_exclusion, useNA = "ifany"))

n_before_cvd <- nrow(df_cohort)

df_cohort <- df_cohort %>%
  filter(
    cvd_exclusion != 1 | is.na(cvd_exclusion)
  )

cat("\nCVD exclusion applied.\n")
cat("Before:", n_before_cvd, "\n")
cat("After:", nrow(df_cohort), "\n")
cat("Excluded:", n_before_cvd - nrow(df_cohort), "\n")
cat(
  "Exclusion rate (%):",
  round(
    (n_before_cvd - nrow(df_cohort)) /
    n_before_cvd * 100,
    2
  ),
  "\n"
)

cat("\nN by year after CVD exclusion:\n")
print(table(df_cohort$year, useNA = "ifany"))

# 32093 -> 23418
# n = 8675 excluded

# =====
# Predictor / covariate cleaning
# =====

# -----
# 8. Stress perception: BP1
# -----
# Original:
# 1 = 매우 많이 느낌
# 2 = 많이 느낌
# 3 = 조금 느낌
# 4 = 거의 느끼지 않음
#
# Reverse:
# Higher value = greater perceived stress

df_cohort <- df_cohort %>%
  mutate(
    BP1_numeric = valid_values(
      BP1,
      c(1, 2, 3, 4)
    ),

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    BP1_reversed = case_when(
      BP1_numeric == 1 ~ 4,
      BP1_numeric == 2 ~ 3,
      BP1_numeric == 3 ~ 2,
      BP1_numeric == 4 ~ 1,
      TRUE ~ NA_real_
    )
  )

cat("\nBP1 reversed check:\n")
print(table(df_cohort$BP1_reversed, useNA = "ifany"))

df_cohort <- df_cohort %>%
  select(
    -any_of(c(
      "BP1",
      "BP1_numeric"
    ))
  )

# -----
# 9. Physical activity
# -----
# Weekly physical activity is converted to MET-min/week.
#
# Vigorous activity:
# 8 MET × minutes/week
#
# Moderate activity:
# 4 MET × minutes/week
#
# Transport activity:
# 4 MET × minutes/week
#
# Final group:
# <600 MET-min/week
# >=600 MET-min/week

clean_activity_domain <- function(
  data,
  participation_col,
  days_col,
  hours_col,
  minutes_col,
  output_col
) {

  p_clean <- paste0(participation_col, "_clean")
  d_clean <- paste0(days_col, "_clean")
  h_clean <- paste0(hours_col, "_clean")
  m_clean <- paste0(minutes_col, "_clean")

  data %>%
    mutate(

      "{p_clean}" :=
        valid_values(
          .data[[participation_col]],
          c(1, 2)
        ),

      "{d_clean}" :=
        valid_range(

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      .data[[days_col]],
      1,
      7
    ),

    "{h_clean}" :=
      valid_range(
        .data[[hours_col]],
        0,
        24
      ),

    "{m_clean}" :=
      valid_range(
        .data[[minutes_col]],
        0,
        59
      ),

    daily_minutes_temp =
      .data[[h_clean]] * 60 +
      .data[[m_clean]],

    "{output_col}" := case_when(

      .data[[p_clean]] == 2 ~ 0,

      .data[[p_clean]] == 1 &
        !is.na(.data[[d_clean]]) &
        !is.na(daily_minutes_temp) &
        daily_minutes_temp <= 1440 ~
        .data[[d_clean]] * daily_minutes_temp,

      TRUE ~ NA_real_
    )
  ) %>%
  select(-daily_minutes_temp)
}

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df_cohort <- df_cohort %>%
```

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  clean_activity_domain(
    "BE3_71",
    "BE3_72",
    "BE3_73",
    "BE3_74",
    "vigorous_work_min_week"
  ) %>%

```

```

  clean_activity_domain(
    "BE3_75",
    "BE3_76",
    "BE3_77",
    "BE3_78",
    "vigorous_leisure_min_week"
  ) %>%

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  clean_activity_domain(
    "BE3_81",
    "BE3_82",
    "BE3_83",
    "BE3_84",
    "moderate_work_min_week"
  ) %>%

```

```

clean_activity_domain(
  "BE3_85",
  "BE3_86",
  "BE3_87",
  "BE3_88",
  "moderate_leisure_min_week"
) %>%

clean_activity_domain(
  "BE3_91",
  "BE3_92",
  "BE3_93",
  "BE3_94",
  "transport_min_week"
) %>%

mutate(

  # -----
  # MET-min/week
  # -----

  total_met_min_week = if_else(

    !is.na(vigorous_work_min_week) &
    !is.na(vigorous_leisure_min_week) &
    !is.na(moderate_work_min_week) &
    !is.na(moderate_leisure_min_week) &
    !is.na(transport_min_week),

    8 * vigorous_work_min_week +
    8 * vigorous_leisure_min_week +
    4 * moderate_work_min_week +
    4 * moderate_leisure_min_week +
    4 * transport_min_week,

    NA_real_
  ),

  # -----
  # 600 MET-min/week cutoff
  # -----

  physical_activity_group = case_when(

    is.na(total_met_min_week) ~ NA_character_,

    total_met_min_week < 600 ~
      "<600 MET-min/week",

    total_met_min_week >= 600 ~
      ">=600 MET-min/week"
  ),

  physical_activity_group = factor(
    physical_activity_group,
    levels = c(
      "<600 MET-min/week",
      ">=600 MET-min/week"
    )
  )
)

```

```

cat("\nPhysical activity MET check:\n")
print(summary(df_cohort$total_met_min_week))

cat(
  "Missing total_met_min_week:",
  sum(is.na(df_cohort$total_met_min_week)),
  "\n"
)

cat("\nPhysical activity group:\n")
print(
  table(
    df_cohort$physical_activity_group,
    useNA = "ifany"
  )
)

df_cohort <- df_cohort %>%
  select(
    -any_of(c(
      "BE3_71", "BE3_72", "BE3_73", "BE3_74",
      "BE3_75", "BE3_76", "BE3_77", "BE3_78",
      "BE3_81", "BE3_82", "BE3_83", "BE3_84",
      "BE3_85", "BE3_86", "BE3_87", "BE3_88",
      "BE3_91", "BE3_92", "BE3_93", "BE3_94",

      "BE3_71_clean", "BE3_72_clean",
      "BE3_73_clean", "BE3_74_clean",

      "BE3_75_clean", "BE3_76_clean",
      "BE3_77_clean", "BE3_78_clean",

      "BE3_81_clean", "BE3_82_clean",
      "BE3_83_clean", "BE3_84_clean",

      "BE3_85_clean", "BE3_86_clean",
      "BE3_87_clean", "BE3_88_clean",

      "BE3_91_clean", "BE3_92_clean",
      "BE3_93_clean", "BE3_94_clean",

      "vigorous_work_min_week",
      "vigorous_leisure_min_week",
      "moderate_work_min_week",
      "moderate_leisure_min_week",
      "transport_min_week"
    ))
  )

# -----
# 10. Sleep duration
# -----
# 2014:
# BP8 = average sleep hours
#
# 2020 / 2022:
# BP16_1 = weekday sleep
# BP16_2 = weekend sleep
#
# 2016 / 2018 / 2024:
# clock-based bedtime/wake time
#

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```
# Weighted average:
# (weekday × 5 + weekend × 2) / 7
```

```
average_sleep_years <- c(2014)
```

```
direct_sleep_years <- c(
  2020,
  2022
)
```

```
clock_sleep_years <- c(
  2016,
  2018,
  2024
)
```

```
all_sleep_years <- c(
  2014,
  2016,
  2018,
  2020,
  2022,
  2024
)
```

```
clock_total_minutes <- function(hour, minute) {
```

```
  hour <- remove_missing_codes(hour)
  minute <- remove_missing_codes(minute)
```

```
  valid_time <-
    hour >= 1 &
    hour <= 24 &
    minute >= 0 &
    minute <= 59
```

```
  hour <- if_else(
    valid_time,
    hour,
    NA_real_
  )
```

```
  minute <- if_else(
    valid_time,
    minute,
    NA_real_
  )
```

```
  hour <- if_else(
    hour == 24,
    0,
    hour
  )
```

```
  hour * 60 + minute
```

```
}
```

```
sleep_duration_from_clock <- function(
```

```
  bed_hour,
  bed_minute,
  wake_hour,
  wake_minute
```

```

) {

  bed_total <-
    clock_total_minutes(
      bed_hour,
      bed_minute
    )

  wake_total <-
    clock_total_minutes(
      wake_hour,
      wake_minute
    )

  duration <-
    wake_total - bed_total

  duration <- if_else(
    wake_total > bed_total,
    duration,
    wake_total + 1440 - bed_total
  )

  duration <- if_else(
    !is.na(bed_total) &
    !is.na(wake_total),
    duration,
    NA_real_
  )

  duration / 60
}

df_cohort <- df_cohort %>%
  mutate(
    weekday_sleep_hours = NA_real_,
    weekend_sleep_hours = NA_real_
  )

# Direct sleep years: 2020 / 2022
df_cohort <- df_cohort %>%
  mutate(

    weekday_sleep_hours = if_else(
      year %in% direct_sleep_years,
      remove_missing_codes(BP16_1),
      weekday_sleep_hours
    ),

    weekend_sleep_hours = if_else(
      year %in% direct_sleep_years,
      remove_missing_codes(BP16_2),
      weekend_sleep_hours
    )
  )

# Clock sleep years: 2016 / 2018 / 2024
df_cohort <- df_cohort %>%
  mutate(

    weekday_sleep_clock =
      sleep_duration_from_clock(

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      BP16_11,
      BP16_12,
      BP16_13,
      BP16_14
    ),

    weekend_sleep_clock =
      sleep_duration_from_clock(
        BP16_21,
        BP16_22,
        BP16_23,
        BP16_24
      ),

    weekday_sleep_hours = if_else(
      year %in% clock_sleep_years,
      weekday_sleep_clock,
      weekday_sleep_hours
    ),

    weekend_sleep_hours = if_else(
      year %in% clock_sleep_years,
      weekend_sleep_clock,
      weekend_sleep_hours
    )
  )
)

df_cohort <- df_cohort %>%
  mutate(

    sleep_avg_weighted =
      (
        weekday_sleep_hours * 5 +
        weekend_sleep_hours * 2
      ) / 7,

    # 2014
    sleep_avg_weighted = if_else(
      year %in% average_sleep_years,
      remove_missing_codes(BP8),
      sleep_avg_weighted
    ),

    # Flag only
    sleep_implausible =
      !is.na(sleep_avg_weighted) &
      (
        sleep_avg_weighted < 3 |
        sleep_avg_weighted > 14
      )
  )

cat("\nSleep duration check:\n")

print(
  summary(
    df_cohort$sleep_avg_weighted
  )
)

print(
  table(
    df_cohort$sleep_implausible,

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    useNA = "ifany"
  )
)

cat("\nSleep duration missing counts by year:\n")

print(
  df_cohort %>%
    group_by(year) %>%
    summarise(
      n = n(),
      sleep_missing =
        sum(
          is.na(
            sleep_avg_weighted
          )
        ),
      sleep_implausible_n =
        sum(
          sleep_implausible,
          na.rm = TRUE
        ),
      .groups = "drop"
    )
)

df_cohort <- df_cohort %>%
  select(
    -any_of(c(
      "BP8",
      "BP16_1",
      "BP16_2",
      "BP16_11",
      "BP16_12",
      "BP16_13",
      "BP16_14",
      "BP16_21",
      "BP16_22",
      "BP16_23",
      "BP16_24",
      "weekday_sleep_hours",
      "weekend_sleep_hours",
      "weekday_sleep_clock",
      "weekend_sleep_clock"
    ))
  )

# -----
# 11. Demographic / socioeconomic / behavior covariates
# -----

df_cohort <- df_cohort %>%
  mutate(
    # Sex
    # 1 = Male

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# 2 = Female
sex_numeric =
  valid_values(
    sex,
    c(1, 2)
  ),

# Household income
# 1 = Low
# 2 = Lower-middle
# 3 = Upper-middle
# 4 = High
household_income =
  valid_values(
    ho_incm,
    c(1, 2, 3, 4)
  ),

# Education
# 1 = Elementary or less
# 2 = Middle school
# 3 = High school
# 4 = College or above
edu_numeric =
  valid_values(
    edu,
    c(1, 2, 3, 4)
  ),

# Marital status
# 1 = Married
# 2 = Unmarried
marri_1_numeric =
  valid_values(
    marri_1,
    c(1, 2)
  ),

# Working status
# 1 = Yes
# 2 = No
employment_status =
  valid_values(
    EC1_1,
    c(1, 2)
  ),

# Body mass index
BMI_numeric =
  valid_range(
    HE_BMI,
    10,
    80
  ),

BMI_group = case_when(
  BMI_numeric < 23 ~ "<23 kg/m2",
  BMI_numeric >= 23 & BMI_numeric < 25 ~ "23-24.9 kg/m2",
  BMI_numeric >= 25 ~ ">=25 kg/m2",
  TRUE ~ NA_character_
),

BMI_group = factor(
  BMI_group,
  levels = c(

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    "<23 kg/m2",
    "23-24.9 kg/m2",
    ">=25 kg/m2"
  )
),

# Obesity category
HE_obe_numeric =
  valid_values(
    HE_obe,
    c(1, 2, 3, 4, 5, 6)
  ),

# Smoking status
# 1 = Never
# 2 = Previous
# 3 = Current
smoking_history = case_when(

  valid_values(
    BS1_1,
    c(1, 2, 3)
  ) == 3 ~ 1,

  valid_values(
    BS3_1,
    c(1, 2, 3)
  ) == 3 ~ 2,

  valid_values(
    BS3_1,
    c(1, 2, 3)
  ) %in% c(1, 2) ~ 3,

  TRUE ~ NA_real_
),

# Alcohol drinking status
# 1 = Never
# 2 = Previous
# 3 = Current
BD1_11_numeric =
  case_when(
    valid_values(BD1, c(1, 2)) == 1 ~ 1,
    valid_values(BD1, c(1, 2)) == 2 &
      valid_values(BD1_11, c(1, 2, 3, 4, 5, 6)) == 1 ~ 2,
    valid_values(BD1, c(1, 2)) == 2 &
      valid_values(BD1_11, c(1, 2, 3, 4, 5, 6)) %in% c(2, 3, 4, 5, 6) ~ 3,
    TRUE ~ NA_real_
  )
) %>%
select(
  -any_of(c(
    "sex",
    "ho_incm",
    "edu",
    "marri_1",
    "EC1_1",
    "HE_BMI",
    "HE_obe",
    "BS1_1",
    "BS3_1",
    "BD1",
    "BD1_11"
  ))
)
```

```

)

cat("\nCovariates check:\n")

print(
  table(
    df_cohort$sex_numeric,
    useNA = "ifany"
  )
)

print(
  table(
    df_cohort$household_income,
    useNA = "ifany"
  )
)

print(
  table(
    df_cohort$edu_numeric,
    useNA = "ifany"
  )
)

print(
  table(
    df_cohort$marri_1_numeric,
    useNA = "ifany"
  )
)

print(
  table(
    df_cohort$employment_status,
    useNA = "ifany"
  )
)

print(summary(df_cohort$BMI_numeric))

print(
  table(
    df_cohort$BMI_group,
    useNA = "ifany"
  )
)

print(
  table(
    df_cohort$HE_obese_numeric,
    useNA = "ifany"
  )
)

print(
  table(
    df_cohort$smoking_history,
    useNA = "ifany"
  )
)

print(
  table(

```

```

    df_cohort$BD1_11_numeric,
    useNA = "ifany"
  )
)

# -----
# 12. Remove temporary variables
# -----

df_cohort <- df_cohort %>%
  select(
    -any_of(c(
      "BS3_1",
      "pulse_abnormal",
      "hypertension_dx",
      "stroke_dx",
      "mi_dx",
      "angina_dx",
      "cvd_exclusion"
    ))
  )

# -----
# 13. Final variable order
# -----

df_cohort <- df_cohort %>%
  select(
    any_of(c(

      "year",
      "ID",

      # Outcome
      "mh_PHQ_S",
      "PHQ_group",

      # Main predictor
      "pulse_rate_60",

      # Behavioral / lifestyle predictors
      "BP1_reversed",
      "total_met_min_week",
      "physical_activity_group",
      "sleep_avg_weighted",
      "sleep_implausible",

      # Demographics / covariates
      "age_numeric",
      "sex_numeric",
      "household_income",
      "edu_numeric",
      "marri_1_numeric",
      "employment_status",
      "BMI_numeric",
      "BMI_group",
      "HE_obese_numeric",
      "smoking_history",
      "BD1_11_numeric"
    ))
  )

```

```

cat("\nFinal variables kept:\n")
print(names(df_cohort))

cat(
  "\nFinal df_cohort shape before predictor missing-value handling:",
  dim(df_cohort),
  "\n"
)

# -----
# 14. Final checks
# -----

cat("\nPHQ group:\n")
print(
  table(
    df_cohort$PHQ_group,
    useNA = "ifany"
  )
)

cat("\nPhysical activity group:\n")
print(
  table(
    df_cohort$physical_activity_group,
    useNA = "ifany"
  )
)

cat("\nMissing values:\n")

print(
  colSums(
    is.na(df_cohort)
  )
)

# -----
# 15. Save cleaned cohort
# -----

write_csv(
  df_cohort,
  "df_cohort_cleaned.csv"
)

cat(
  "\nSaved file: df_cohort_cleaned.csv\n"
)

print(table(df_cohort$PHQ_group, useNA = "ifany"))

```